Mikayla Deehring

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Ph.D. Student — Electrical & Computer Engineering

Research Summary

My research leverages wearable sensors and interpretable machine learning to develop personalizable platforms for at-home rehabilitation and remote patient monitoring that improve health outcomes and promote equity in healthcare.

Education

2025–Present **Ph.D., Electrical & Computer Engineering**, *Rice University*, Houston, TX, GPA: 3.93
Area: HCI, Machine Learning, Remote Patient Monitoring, Rehabilitation.
Advisor: Dr. Momona Yamagami

2023–2025 **M.S., Electrical & Computer Engineering**, *Rice University*, Houston, TX, GPA: 3.91
Area: HCI, Machine Learning, Remote Patient Monitoring.
Advisor: Dr. Momona Yamagami

2019–2023 **B.S., Biomedical Engineering**, *University of Houston*, Houston, TX, GPA: 3.89
University Honors, Dean's List, Summa Cum Laude

Research Experience

2023–Present **Graduate Research Assistant**, *Yamagami Lab, Rice University*, Houston, TX

Exergaming for Pediatric Stroke Survivors:
- Develop and evaluate an end-to-end platform for sensor- and exercise-agnostic exergaming using user-centered design that benefits the understudied population of pediatric stroke survivors.

Biosignals for Rehabilitation:
- Use data collected from commercially available smartwatches to investigate the relationship between biosignals and exertion, and remotely track adherence to rehabilitation.

2021–2023 **Undergraduate Research Assistant**, *CNEL Lab, University of Houston*, Houston, TX
Advisor: Dr. Nuri Ince
Research Focus: Developing an accelerometer-based platform to monitor tremors in Parkinson's Disease patients.

Summer 2020 **HERE Scholar**, *Houston Early Research Experience*, University of Houston, Houston, TX
Advisor: Dr. Amin Kiaghadi
Focus: Urbanism in Houston. Completed intensive cohort-based training on research methodologies and proposal development.

Publications & Presentations

Published Proceedings

[4] M. E. Deehring, H. P. Katzman, A. Dhala, and M. Yamagami, "Data-driven modeling to predict prehabilitation adherence in frail individuals with ESKD," in *RESNA Annual Conference Online Proceedings*, Chicago, IL, 2025.

Conference Presentations

[2] M. E. Deehring, A. Clearman, S. Fraser, and M. Yamagami, "Press Play on Rehab: A Pilot Study of Exergaming for Pediatric Stroke," in *Summer School on Neurorehabilitation*, Baiona, Spain, 2026.

Poster Presentations

[3] M. E. Deehring, H. P. Katzman, S. Dalmia, A. Clearman, S. Fraser, and M. Yamagami, "Pilot Study for a Pediatric Stroke Exergaming System," in *IEEE BSN*, Los Angeles, CA, 2025.

[5] M. E. Deehring, A. Dhala, and M. Yamagami, "Predicting Adherence to At-Home Rehabilitation Using Biosignals," in *IEEE BSN*, Chicago, IL, 2024.

Manuscripts in Preparation

[1] M. E. Deehring, E. C. Lorenz, H. P. Katzman, R. Isac, F. Sasangohar, A. Dhala, and M. Yamagami, "Wearable Sensors Predict Exertion Without Changes in Heart Rate for ESKD Patients."

Honors & Awards

2026 UTHealth/MD Anderson Cancer Center CCTS T32 Fellowship

2022 University of Houston Biomedical Engineering Inspiration Award

2022 University of Houston Undergraduate Biomedical Engineering Junior of the Year

2019–2023 University of Houston Dean's List

2019–2023 Academic Excellence Scholarship

Internships

2022–2023 **Research and Design Intern**, *Starling Medical*, Medical Device Startup
Mentor: Fergus Wong
- Designed and prototyped an at-home uroflowmetry device to monitor patients with benign prostatic hypertrophy and other conditions causing lower urinary tract symptoms.
- Designed and executed a study to test the ability of a urinalysis device to measure the concentration of multiple analytes in solution based on plausible biological ranges.

Teaching

2024, 2025 **Teaching Assistant**, *Neural Interface Engineering Lab*, Rice University, Houston, TX
- Administer weekly lab with experiments related to biosignal recording and analysis.

Mentoring

Undergraduate Students

Spring 2026 – Present **Marco Montalbano**, *Rice University*
Project: Implementation of exergaming prototype for pediatric stroke survivors.

Spring 2026 **Ethan Harjabrata**, *Rice University*
Project: Implementation of exergaming prototype for pediatric stroke survivors.

Fall 2025 – Spring 2026 **Sanjana Dabbiru**, *Rice University*
Project: Implementation of exergaming prototype for pediatric stroke survivors.

Summer 2025 – Spring 2026 **Joyce Wu**, *Rice University*
Project: Comparing marker-based motion capture to markerless motion capture with pediatric stroke survivors.

Summer 2025 **Shivam Dalmia**, *Rice University*
Project: Implementing sensor-agnostic gesture recognition on RGB-D data to classify movement types during occupational therapy and exergaming for pediatric stroke survivors.

2024 – 2025 **Philo Katzman**, *Rice University*
Project: Researching how biosignals can be used to predict exertion during rehabilitation exercise.

Summer 2024 **Tony Martinez**, *Rice University*

Project: Creating a VR application in Unity to measure visual and motor attention. Concurrently awarded 2024 Rice University Summer Undergraduate Research Fellowship.